EDUGENIE – PROJECT DEVELOPMENT PHASES

2. Requirement Analysis Phase

The requirement analysis phase identifies what the EduGenie system needs to perform successfully. Both functional and non-functional requirements were considered.

Functional Requirements

The system should provide the following major functionalities:

1. Question and Answer

Users can enter an academic or general question. EduGenie sends the question to the AI model and displays an appropriate answer.

2. Concept Explanation

Users can enter a difficult topic and request a simplified explanation.

3. Quiz Generation

The system generates multiple-choice questions from a topic or passage.

4. Text Summarization

EduGenie summarizes lengthy educational content into shorter and easier-to-understand content.

5. Learning Recommendations

The system generates a structured learning path from beginner to advanced level.

6. Web Interface

Users should be able to access all features through a simple browser-based interface.

Non-Functional Requirements

- Simple and easy-to-use interface.
- Fast response time.
- Responsive design.
- Secure API-key handling.
- Modular software architecture.
- Reliable error handling.
- Compatibility with common operating systems.
- Ability to run on systems with limited hardware resources.

Hardware Requirements

- Computer or laptop.
- Minimum 4 GB RAM recommended.
- Internet connection.
- Modern web browser.

Software Requirements

- Python 3.10+
- FastAPI
- Uvicorn

- Jinja2
- HTML5
- CSS3
- Google Gemini API

Required Python libraries